REQUIREMENTS TRACEABILITY MATRIX

Requirement ID	Requirement Description	Type	Priority	Design / Component	Implementation Evidence	Test Case ID	Test Result	Final Status
FR-01	Sign in with email and password and go to the dashboard for the user's role.	Functional	Must	UC-01; sign-in page; auth Server Action	src/app/actions/auth.ts (loginAction); src/lib/password.ts; src/lib/session.ts	TC-001, TC-008, TC-011, TC-013, TC-110, TC-111, TC-112	Pass	Met
FR-02	Student sign-up with a CTU student email and a strong password.	Functional	Should	UC-01; sign-up page	src/app/actions/auth.ts (signupAction); src/lib/signup-validation.ts	TC-113, TC-114, TC-115, TC-116	Pass	Met
FR-03	Restrict each page and action to the roles allowed to use it.	Functional	Must	Access control layer (proxy, layouts, page-level role checks, session)	src/proxy.ts; src/lib/session.ts (requireRole on every protected page and layout); ownership checks in actions	TC-112, TC-117, TC-118, TC-119, TC-120, TC-120b, TC-121, TC-154, TC-168, TC-171	Pass	Met
FR-04	Create a daily logbook entry (date, description, hours 0-24, mentor email).	Functional	Must	UC-02; logbook_entries table	src/app/actions/logbook.ts (createLogbookEntryAction); src/components/new-entry-form.tsx	TC-005, TC-006, TC-012, TC-014, TC-122, TC-123	Pass	Met
FR-05	Attach one optional evidence file (PDF, PNG, JPG, WEBP).	Functional	Should	UC-03; data/uploads	src/app/actions/logbook.ts (extension allow-list, random file names)	TC-124, TC-125	Pass	Met
FR-06	Submit a Draft or Rejected entry for verification; 7-day one-time link; status Pending.	Functional	Must	UC-04; verification_tokens table; sequence diagram steps 1-5	src/app/actions/logbook.ts (submitEntryForVerificationAction); repository.submitEntryForVerification (transaction)	TC-127, TC-128, TC-129, TC-137	Pass	Met
FR-07	Mentor opens the link without an account and approves or rejects the entry.	Functional	Must	UC-05; mentor verification page	src/app/verify/[token]/page.tsx; src/app/actions/verify.ts	TC-130, TC-131, TC-132	Pass	Met
FR-08	A link works only once and only before it expires.	Functional	Must	Sequence diagram alt frame; activity diagram	src/app/actions/verify.ts; repository.recordMentorDecision (WHERE used_at IS NULL)	TC-133, TC-134, TC-135, TC-136, TC-137	Pass	Met
FR-09	Show each entry's status (Draft, Pending verification, Verified, Rejected).	Functional	Must	Status badges	src/components/status-badge.tsx; src/lib/logbook-metrics.ts	TC-003, TC-004, TC-138	Pass	Met
FR-10	Student views each placement's entries, newest first, and its verified hours against its required hours.	Functional	Must	Student logbook page	src/app/student/page.tsx; src/app/student/placements/[id]/page.tsx; sumVerifiedHours	TC-002, TC-006, TC-132, TC-138, TC-155	Pass	Met
FR-11	Coordinators and lecturers see each student's status for their current placement, verified hours, entries, pending count and missing documents.	Functional	Must	UC-06; coordinator dashboard	src/app/coordinator/page.tsx; studentOverviewStatus, currentPlacement	TC-009, TC-139, TC-170, TC-174	Pass	Met
FR-12	Record created, updated, verified and link-used times automatically.	Functional	Should	Timestamp columns in the ERD	src/lib/repository.ts; database/schema.sql	TC-131, TC-140	Pass	Met
FR-13	Evidence files only downloadable by the owner, coordinators and lecturers.	Functional	Must	Evidence route	src/app/api/evidence/[fileName]/route.ts	TC-126	Pass	Met

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FR-14	Sign out ends the session.	Functional	Must	UC-07; header	src/app/actions/auth.ts (logoutAction); src/components/app-header.tsx	TC-007, TC-010, TC-141, TC-146	Pass	Met
FR-15	Add WIL placements, each with its own required hours; every entry belongs to a placement.	Functional	Must	UC-08; wil_placements table; placement pages	src/app/actions/placements.ts; src/app/student/placements; src/lib/repository.ts	TC-150, TC-151, TC-152, TC-153, TC-154, TC-155, TC-156, TC-174	Pass	Met
FR-16	Upload WIL documents: MOU, Commencement Letter and Code of Conduct per placement; Attendance Register per semester.	Functional	Must	UC-09; placement_documents table; documents section	src/app/actions/documents.ts; src/lib/documents.ts; src/app/api/documents/[id]/route.ts	TC-157, TC-158, TC-159, TC-160, TC-161, TC-162, TC-163, TC-164, TC-165	Pass	Met
FR-17	Coordinators and lecturers open a student's profile with placements, documents and logbook.	Functional	Must	UC-10; student profile page	src/app/coordinator/students/[id]/page.tsx	TC-166, TC-167, TC-168, TC-169, TC-170	Pass	Met
FR-18	A placement is WIL completed when its verified hours reach its required hours, and is then locked.	Functional	Must	Status rules; locks in the Server Actions	src/lib/logbook-metrics.ts (placementCompletion, studentOverviewStatus); checks in src/app/actions	TC-172, TC-173, TC-174	Pass	Met
NFR-01	A first-time student creates and submits an entry in under 3 minutes.	Non-Functional	Should	Four-field entry form; Copy link button	src/components/new-entry-form.tsx	TC-149	Not Tested	Partially Met
NFR-02	Pages and actions respond within 2 seconds.	Non-Functional	Should	SQLite with indexes; server rendering	src/lib/db.ts; database/schema.sql (indexes)	TC-144	Pass	Met
NFR-03	Salted password hashes, HttpOnly signed session cookie, 128+ bit tokens, server-side checks.	Non-Functional	Must	Security design (Chapter 4.7)	src/lib/password.ts; src/lib/session.ts; src/app/actions/logbook.ts (randomBytes(24))	TC-121, TC-126, TC-127, TC-142, TC-160, TC-162, TC-171	Pass	Met
NFR-04	Database constraints reject invalid data; multi-step updates are atomic.	Non-Functional	Must	ERD constraints; transactions	database/schema.sql; src/lib/db.ts (inTransaction)	TC-136, TC-143	Pass	Met
NFR-05	TypeScript, layered modules, lint passes, README set-up in five commands or fewer.	Non-Functional	Should	Three-layer architecture	Source structure; README.md	TC-147	Pass	Met
NFR-06	Works in current Chrome and Edge; usable at 375 px and 1280 px.	Non-Functional	Should	Responsive Tailwind layout	src/app/*/layout.tsx; overflow-x-auto tables	TC-145, TC-146	Pass	Met
NFR-07	Only data needed for WIL tracking is stored; synthetic data only until consent.	Non-Functional	Must	Data minimisation; Ethical Clearance	database/seed.sql; database/demo-documents	TC-148	Pass	Met